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# The use of scenario tree models in support of animal health surveillance: A scoping review

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ABSTRACT

*Background:* Scenario tree modelling is a well-known method used to evaluate the confidence of freedom from infection or to assess the sensitivity of a surveillance system in detecting an infection at a certain design prevalence. It facilitates the use of data from various sources and the inclusion of risk factors into calculations, while still obtaining quantitative estimates of surveillance sensitivity and probability of freedom.
*Objectives:* We conducted a scoping review to identify scenario tree models (STMs) applied to assess freedom from infection in veterinary medicine, characterize their use, parameterisation, reporting and potential limitations.
*Eligibility criteria:* We included published scientific articles and grey literature that were a) neither reviews nor expert opinions, b) aimed to assess freedom from infection, provided methods to assess it, or aimed to estimate the sensitivity of a surveillance program for early detection of an infection at a design prevalence, c) targeted infection in animals and d) used scenario tree modelling. The search covered documents published between January 2006 and August 2021.
*Design:* Several search methods were used to retrieve scientific articles and grey literature relevant to the subject. The search strategy included searching in scientific databases and/or grey literature repositories, contacting experts across the world that previously worked with STMs and retrieving citations from relevant reviews.
*Results and discussion:* Four hundred twenty-four distinct documents were retrieved with our search string. After screening, data was extracted from 99 documents representing 67 projects. Forty different animal diseases were modelled with STMs, the most represented being infections with tuberculous *Mycobacterium* sp., Avian Influenza A virus and *Brucella* sp. STMs were mostly used for diseases of cattle, swine and wild mammals. Results showed that STMs were used in a large variety of studies, are very versatile and were used in disparate frameworks.

*Abbreviations:* STM, Scenario tree model; WTO, World Trade Organization; WOAH, World Organization of Animal Health formerly known as OIE.

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However, we also found that studies are not reported in a standardized way and often lack important information. This makes results hard to interpret, compare and reproduce. Additionally, we identified common assumptions and misconceptions, the most important ones regarding sensitivity and specificity, which could have an impact on the results of the studies using STMs.
*Conclusion:* We recommend the elaboration of internationally agreed guidelines about how to report results from STMs in a uniform manner. Such guidelines should include information on the study setting, procedures and analyses, but also on how the results could be interpreted concerning freedom from infection.

**Definitions**

In this article, many technical terms related to animal health surveillance and to the scenario tree model methodology are used. Terms deriving from the scenario tree model methodology follow the definitions by Martin et al. (2007b). Terms related to animal surveillance are derived from the Animal Health Surveillance Terminology Final Report from Pre-ICAHS Workshop (Hoinville, 2013).

## 1. Introduction

The World Trade Organization (WTO) requires in the Sanitary and Phytosanitary agreement that member states follow specific rules to introduce or maintain sanitary measures to ensure safe trade and to prevent unjustified trade restrictions. These measures should be "based on scientific principles and […] not maintained without sufficient scientific evidence" (WTO, 1995). The World Organisation for Animal Health WOAH (2024) has developed general guidelines to demonstrate freedom from disease that among others include requirements to the veterinary services, legislation, and the performance of surveillance for the infectious agent (WOAH, 2024). To provide the required scientific evidence, member states regularly assess the confidence of freedom from infection for many different pathogens in their country. Historically, the surveillance for the pathogenic agent was done either by representative random sampling of the target population, which was costly and resource-intensive (Cameron et al., 2014; Cameron, 2012; Cameron and Baldock, 1998; Cannon and Roe, 1982; Doherr and Audigé, 2001), or by expert elicitation, which was subjective and could not provide high levels of confidence (Cameron et al., 2014).

Between the end of the 20th and the beginning of the 21st century, several methods were proposed to minimize the difficulties related to the sampling process. These included optimising the size of the sample pool necessary to assess freedom from infection with a chosen level of confidence—e.g. two-stage sampling (Cameron and Baldock, 1998)—or increasing confidence by considering historical surveillance data in freedom from infection calculations (Hadorn et al., 2002; Schlosser and Ebel, 2001).

Martin et al. (2007b) developed a new method based on scenario trees to allow for the combination of data from several sources—including non-random surveys. This method delivers quantitative results and permits to achieve a higher sensitivity by making the surveillance effort depend on the sum of the hypothesized probability of infection in different strata of the population, i.e., risk-based sampling. These strata are based on known risk factors of infection. This method also allows to sample at several stages and to assess the sensitivity of several detection methods in series. The ability to combine data from several sources was a turning point, as assessments of freedom from infection could now include all data sources in addition to random sampling, such as risk-based sampling or passive surveillance data.

Since then, the method has been used not only for research purposes but also routinely by countries or regions to assess their status of freedom from infection. At the time of writing, the original description of the scenario tree method by Martin et al. (2007b) has been cited 258 times (based on Web of Science Core Collection statistics, last retrieved 2024–02–26). It is the method that the Food and Agriculture Organization of the United Nations uses in its manual for risk-based disease surveillance (Cameron et al., 2014). In addition, it was used to assess the effectiveness of early detection (Welby et al., 2017) and to monitor the sensitivity of surveillance systems for various infections, including endemic diseases (Hadorn et al., 2008; Hadorn and Stärk, 2008; Hernández-Jover et al., 2011; Huneau-Salaün et al., 2015). The method has also generated interest in other fields, like monitoring of animal welfare and of antimicrobial residues in farmed animals, as well as being used in human public health and in the management of invasive animal species (Alban et al., 2016; Dominiak et al., 2011; Hannunen and Tuomola, 2020; Huneau-Salaün et al., 2015; Michael et al., 2018).

Several tools applying various software were developed to help use scenario tree models (STMs). EpiTools (Sergeant 2018, RRID: SCR_022835), a suite of web tools "intended for use by epidemio-logists and researchers involved in estimating disease prevalence or demonstrating freedom from [infection] through structured surveys" (https://epitools.ausvet.com.au, accessed 2024–02–26) was developed by Ausvet to enable, among other options, the use of STMs. In R (R Core Team 2023, RRID:SCR_001905), two distinct packages integrate STMs: *RSurveillance* (Sergeant 2020, RRID:SCR_021674) and *freedom* (Rosendal (2020), RRID:SCR_021675). Functions from *RSurveillance* were also merged into the *epiR* package (Stevenson and Seargeant (2024), RRID: SCR_021673) in 2020 (https://cran.r-project.org/web/packages/epiR/NEWS, accessed 2024–02–26) and is now maintained there. Other tools were also developed to support the implementation of STMs, as for instance the OptisampleTM web application (Alba et al., 2017). Since Martin et al. (2007b) published the method, STMs have been used for different purposes, and with several addenda or changes to the original methods.

We conducted a scoping review to characterize the use, parameterisation, reporting and potential limitations of STMs used in veterinary medicine to evaluate either the sensitivity of surveillance programs or the probability of freedom from infection.

## 2. Material and methods

This review followed the PRISMA extension for scoping reviews (Tricco et al., 2018) and this manuscript is reported according to its guidelines (PRISMA-ScR checklist available as Supplementary Material 1). A protocol outlining the plan for the review was developed *a priori*, following the PRISMA extension for protocols (Moher et al., 2016). Some adaptations were made given that the PRISMA extension for protocols was created for systematic reviews. All details for search strategy and screening can be found in the protocol (Supplementary Material 2).

In the field of animal disease surveillance, the concepts "Disease freedom" (or "Freedom from disease") and "Freedom from infection" are often used interchangeably. Throughout this text, we will use "Freedom from infection" for both. For the sake of simplicity, we have also used this terminology when applied to freedom from parasitic infestation.

### 2.1. Search strategy

The research questions were formulated within the PICo framework (problem, interest, and context) (Stern et al., 2014) as "a) How are scenario tree models used to assess freedom from infections in animals, b) how are they parametrised, c) how are the results reported, and d) what are the limitations?"

Scientific literature was searched in PubMed, Web of Science and CAB Direct. The search strings are described in the protocol (Supplementary Material 2). The search focused on the combination of two main concepts: "scenario tree" and "freedom from infection in animal"; the latter was split between the "freedom from infection" and the "infection in animal" concepts.

Additionally, from PubMed and Web of Science, we retrieved all the publications citing Martin et al. (2007b) and/or Martin et al. (2007a), which first described the method and its application, respectively.

We also searched for grey literature on the topic through ProQuest and the Networked Digital Library of Theses and Dissertations (NDLTD), using a search string similar to the one used for the scientific literature databases.

Finally, we conducted a search verification process. We reached out to the veterinary epidemiology community through the Epivet mailing list (University of Prince Edward Island, 2023), providing a description of our study and its goals and asking experts to send us relevant documents on the topic.

A table summarising the search procedures can be found as Supplementary Material 3.

### 2.2. Screening

Every document was screened by two independent reviewers. The screening involved a two-step process: a) title and abstract screening; and b) full-text screening. The criteria for inclusion and exclusion are available in the protocol.

We checked for duplicates with EndNote X8 (The EndNote Team, 2013) and thereafter uploaded the material to Covidence (Veritas Health Innovation, n.d.) to manage the screening process.

A calibration exercise was performed before each step of the screening. GD selected the documents based on their different scope, so that the reviewers' agreement could be tested in heterogeneous studies. The results were discussed between all reviewers and the criteria clarified on a frequent basis. The calibration before the title and abstract screening was performed on 13 documents (six journal articles, four thesis and three reports). The calibration before the full text screening was performed on eight documents.

The eligibility criteria considered were the following: a) the study is neither a review nor an expert opinion, b) the document aims to assess freedom from infection, provides methods to assess it, or aims to estimate the sensitivity of a surveillance program for early detection of an infection at a design prevalence, c) the study targets infection in animals, and d) the publication uses scenario tree models. We did not set any language limitation. Adaptations to the eligibility criteria for the title and abstract screening are described in the protocol.

### 2.3. Indexing the projects

To avoid overrepresentation of individual projects in our summary of results (e.g. several documents reporting results from the same study) the documents were grouped into projects. Projects were defined as follows: a project comprises one or more documents that used: a) the same methodology; and b) similar type of data sources to produce similar information for the same herd, region or country (Fig. 1).

Some projects ($n = 6$) were updates of previously published STMs including minor changes either with respect to the methods or to the geographical area covered (Fig. 1). These projects were marked as updates and merged with the original project.

All projects identified were given a unique identifier (Supplementary Material 4).

### 2.4. Data extraction

An extraction tool was created in Microsoft Excel Spreadsheet (RRID: SCR_016137). It was tested and adapted in a calibration exercise that included AM, DF, EM, GD, LPC, MGG and PH using six projects. The extraction tool comprised questions which were grouped into several categories: study characteristics such as objectives, publication form, target population, infection of interest and timeframe; methods, including software used, outline of the scenario tree, model type, surveillance characteristics and changes from the original method; parameters, including sources of data, values used for different parameters, estimation of the confidence of the parameters and of the model; information about the results; and potential limitations of the study. The template of the extraction tool is available as Supplementary Material 5.

We also investigated if the authors assessed the confidence of their inputs into the model. We did not evaluate the quality of the references provided for the values of the parameters. But we recorded the type of source used to inform the parameters and checked if the definition and values attributed were provided for each parameter. We also recorded if the uncertainty in input parameters was considered by attributing probability distributions to the parameters, instead of single values. We also investigated if sensitivity analysis were conducted. Finally, we also created a summary score concerning the confidence of input parameters. We considered as minimum requirements to express them as a probability distribution, or when they were expressed as single values, to make simulations with different values in order to assess their influence on the model. We considered that confidence was: a) fully assessed if all parameters were assessed; b) partially assessed if some of them were missing; and c) not assessed if none was assessed. This analysis was only performed on projects published in scientific journals.

AM, DF, EM, FMS, FS, GD, ISB, LPC, MGG, PH, TK, XK and VHSO participated in the data extraction process. When a project used several different scenario trees—e.g. to compare different surveillance systems—data was extracted separately for each scenario tree.

The reviewers met in a regular basis to discuss questions emerging during the extraction process. During these meetings, the reviewers also discussed their impressions concerning biases of the included studies and methodological flaws related to the use of STM. This culminated in a final meeting between the reviewers in which common flaws in the development and reporting of STMs were collected and discussed. This qualitative information is also presented and discussed in this review.

### 2.5. Data analysis

The data was imported to R *4.3.1* (R Core Team 2023, RRID: SCR_001905) with the package *readxl* (Wickham and Bryan, 2023).

The data was analysed using descriptive statistics. Data transformation and summary statistics were derived using *dplyr* (Wickham et al. 2023b, RRID:SCR_016708) and *tidyr* (Wickham et al. 2024, RRID: SCR_017102). Most of the analysis were done at the project level, with some exceptions for analyses related to the layout of scenario trees, which were done at scenario tree level.

For Fig. 2, coordinates for the worldmap were extracted from the package *rworldmap* (South 2011; RRID:SCR_023394). The *lwgeom* (Pebesma 2024, RRID:SCR_022812), *sp* (Pebesma and Bivand 2005; Bivand et al. 2013, RRID:SCR_021328) and *sf* (Pebesma 2018; Pebesma and Bivand 2023, RRID:SCR_023393) packages were used to transform the coordinates in order to plot the map with the Winkel Tripel projection. The map was created and the data added using the package *ggplot2* (Wickham 2016; Wickham et al. (2023a); RRID:SCR_014601).

## 3. Results

### 3.1. Review process

The number of documents retrieved and selected after each step of the review process can be found in Fig. 1.

We retrieved 278 documents with the search strings through

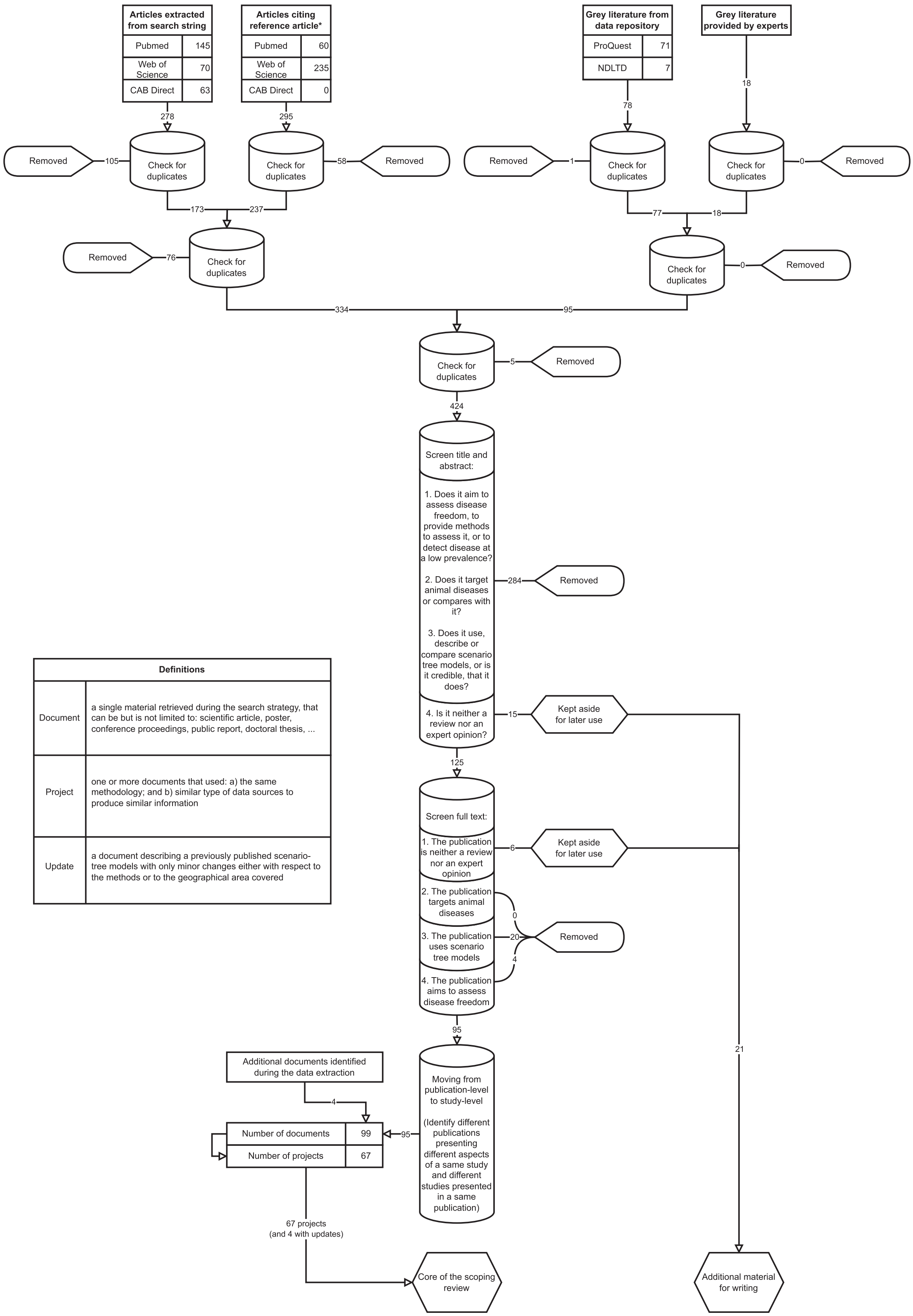


**Fig. 1.** Flowchart of the process of the scoping review.

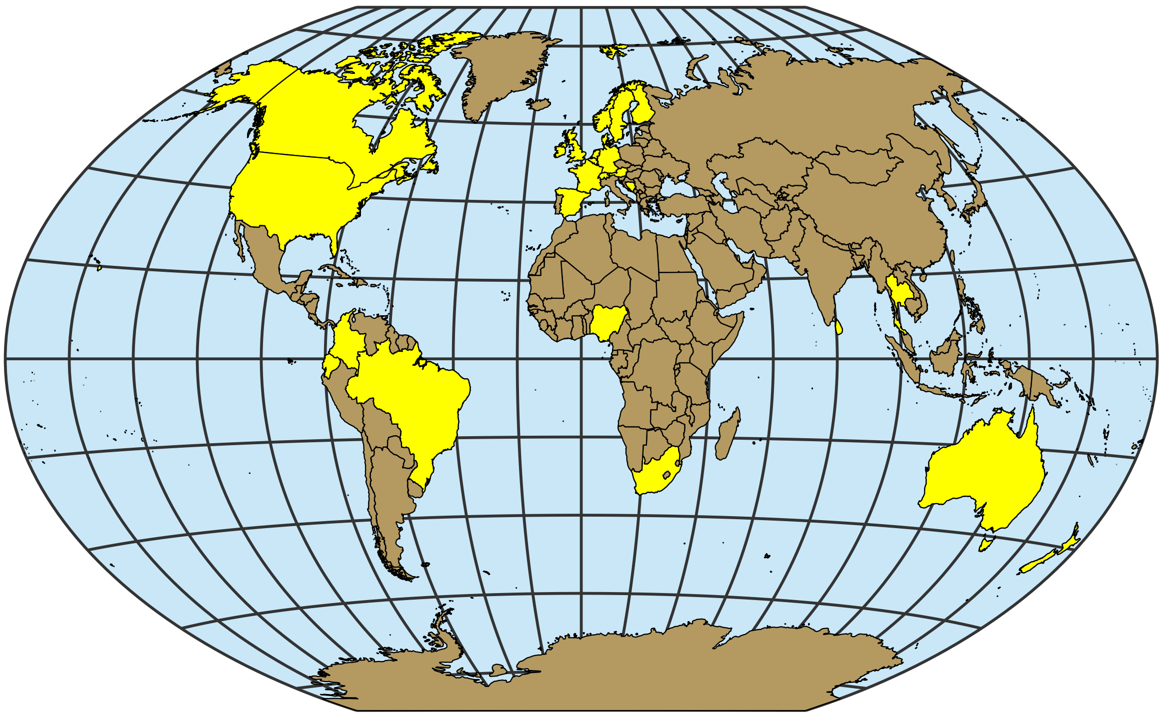

**Fig. 2.** Countries where surveillance programmes for which scenario trees were modelled were in place.

databases of scientific articles. Two hundred ninety-five (295) documents were retrieved from the citation search. Seventy-eight (78) documents were retrieved from grey literature databases, and 18 documents were provided to us by experts during the search verification process. After checking for duplicates, 424 documents were included in the process for title and abstract screening. Of these, 125 were kept for full text screening. After the full text screening, 95 documents remained for the extraction process. During the title and abstract screening, 59 conflicts between the evaluators out of 424 documents (13.9 %) were resolved by mutual agreement. Eighteen conflicts out of 125 documents (14.4 %) were resolved in the same way during the full text screening.

At the end of the screening process, we obtained 95 documents. During the extraction process it was found necessary to include 4 referred papers/reports to obtain necessary data, amounting to a total of 99 documents.

### 3.2. General information on the studies

Data was extracted from a total of 99 documents. This represented a total of 67 projects, originating from 28 countries (Fig. 2). Of these 67 projects, 44 (65.7 %) assessed the probability of freedom from infection, while 23 (34.3 %) assessed only the sensitivity of the surveillance system (without evaluating freedom from infection).

In 31 projects, authors stated that the objective was to assess freedom from infection at a geographical level; 15 projects to allow exportation of animals or animal products, seven projects to be able to impose restrictions on importation, four projects as part of an eradication programme and seven projects to provide data to take decisions concerning inland policies. In three projects, authors wanted to assess freedom from infection at herd level; one project as part of a certification scheme and two projects as part of a control programme. In 49 projects, authors stated that their aim was to assess the sensitivity of the surveillance programmes; 33 projects to compare the sensitivity of different surveillance systems, ten projects to estimate if the surveillance programme met international requirements, eight projects to estimate if the surveillance programme met objectives fixed nationally or locally and five projects that aimed to assess the sensitivity of the surveillance programme for other reasons. In 22 projects, authors stated that they had different aims than the ones listed above.

Most projects used evidence of infection as the case definition (56 projects), while 5 projects used symptoms or pathological findings as case definition. We could not find the information of the case definition for six projects.

Most projects were reported as scientific articles (58/67, 86.6 %). Five (7.5 %) were published as reports, three (4.5 %) as theses, seven (10.4 %) as conference proceedings and five (7.5 %) as posters. Posters could either be retrieved as is, i.e. the whole poster was available to the reviewers, or only through long conference abstracts in proceedings.

Freedom from 40 different infections was analysed. Information on the number of projects that had at least one scenario tree describing a surveillance programme for each infection can be found in Table 1 and in Supplementary Material 6. Table 2 provides the number of projects that included data for each animal category.

Table 3 provides an overview of different general information on the projects and studies. The geographical scale is presented at the project level. However, it is worth mentioning that only one single project used a scenario tree that was, in itself, international. This project from Knight-Jones et al. (2010) was focusing on a specific region around a lake that lies at the border of Germany, Switzerland and Austria. This project had only one scenario tree for the whole region, across the three countries, and we classified it as regional, as we considered the geographic level of interest to be regional though it covered several countries. Other international projects used different scenario trees to describe and compare surveillance systems in different countries. Additionally, when focussed on disease freedom, international projects aimed to calculate the probability of freedom for each country, and not to calculate a global probability of freedom.

### 3.3. Information about the models used

#### 3.3.1. Outline of the scenario tree

Information about the use of different node types in the scenario trees can be found in Table 4.

Eleven projects (16.4 %) used neither different components nor risk

**Table 1**
Number of projects (n = 67) identified in a scoping review (2006–2021) that used scenario tree for each infection, and the regulation of the infection according to the Regulation (EU) 2016/429 on transmissible animal diseases (“Animal Health Law”).

| Infection with | Number of projects | Agent | Categorization according to the Reg (EU) 2016/429* |
|---|---|---|---|
| Tuberculous *Mycobacterium* sp. | 12 | Bacteria | B |
| Influenza A virus (Avian Influenza) | 9 | Virus | A |
| *Brucella* sp. | 6 | Bacteria | B |
| Bluetongue virus | 4 | Virus | C |
| Classical swine fever virus | 3 | Virus | A |
| Bovine leukemia virus | 3 | Virus | C |
| Suid herpesvirus 1 (Aujeszky’s disease) | 3 | Virus | C |
| Porcine reproductive and respiratory syndrome virus | 3 | Virus | D |
| *Mycobacterium avium* subspecies *paratuberculosis* | 3 | Bacteria | E |
| *Trichinella* sp. | 3 | Parasite | Infection with this pathogen not regulated in the Animal Health Law, but regulated in other texts |
| African horse sickness virus | 2 | Virus | A |
| Foot and mouth disease virus | 2 | Virus | A |
| Bovine alphaherpesvirus 1 (infectious bovine rhinotracheitis) | 2 | Virus | C |
| Bovine viral diarrhoea virus | 2 | Virus | C |
| *Echinococcus multilocularis* | 2 | Parasite | C |
| Chronic wasting disease prion | 2 | Prion | Infection with this pathogen not regulated in the Animal Health Law, but regulated in other texts |
| Scrapie prion | 2 | Prion | Infection with this pathogen not regulated in the Animal Health Law, but regulated in other texts |
| Other infections reported in only one project | 23 | - | - |

* Infections can be categorized into several categories according to the European Animal Health Law. For the sake of simplicity, only the higher category was used to classify them. Also, for infections that can have different classifications according to the serotype or the species, such as infection with Influenza A virus or *Salmonella* sp., the highest category was used for this purpose.

**Table 2**
Number of projects (n = 67) identified in a scoping review (2006–2021) that used scenario tree for each animal category.

| Animal category | Number of projects |
|---|---|
| Cattle | 25 |
| Swine | 18 |
| Wild mammals | 12 |
| Poultry | 8 |
| Small ruminants | 7 |
| Equine | 3 |
| Salmonids | 2 |
| Crustaceans | 1 |
| Dogs | 1 |
| Rabbits | 1 |
| Wild birds | 1 |

factors.

Most projects used one or two infection nodes (Table 4), which were mainly animal and/or herd level. Two projects used more than two infection nodes. One of these (Ågren et al., 2018) used both animal and herd level, but also groups of animals as an intermediate infection node between the two. The other one (Welby et al., 2010) reported the use of region status, farm status and animal status as infection nodes. For all the projects, reported infection nodes were variations of animal status, animal group and herd status, or status of geographical regions.

Two types of detection nodes were used. They could either be diagnostic tests, which should generally be well standardised and more objective, and other detection nodes for which we would expect more heterogeneity (e.g. visual inspection, clinical examination). Only three projects (4.5 %) did not use any diagnostic test in at least one scenario tree. Two of these projects (Dadios et al., 2012; Foddai et al., 2015) did not assess freedom from infection, but evaluated the sensitivity of meat inspection. The authors recognized that positive detection at meat inspection would be followed by a laboratory test, but argued that they did not need to model it to achieve their purpose. Another project assessed freedom from infection, but reported that confirmatory diagnosis after meat inspector detection at slaughterhouse would have a sensitivity and specificity of one and therefore did not model it (Hernández-Jover et al., 2011).

Martin et al. (2007b) mentioned in their original article the possibility of overlap between different components and proposed a method to consider this. Overlaps occur when a same unit (e.g. farm, animal) is tested in different components. The assumption of independence between components is then not met and corrections must be made to avoid overestimation of the overall sensitivity. However, on the 11 documents that mentioned that there was an overlap, only six (54.5 %) explicitly mentioned that they considered it in the calculations, and they almost all reported to follow the proposition of Martin et al. (2007b) (Foddai et al., 2020; Frössling et al., 2013, 2009; Stärk et al., 2014; Veldhuis et al., 2017; Wahlström et al., 2010), with only one document mentioning the use of another method (Dadios et al., 2012). Most authors did not report if they checked their model for overlapping components.

Forty-eight projects (71.6 %) provided a graphical representation of the scenario tree model they developed, among which 43 were published in scientific journals.

### 3.3.2. *Sources of data*

Table 4 summarizes model information retrieved from the included projects.

Diverse data originating from a wide variety of either passive or active surveillance programmes were used (Table 3). Data could originate from healthy, asymptomatic animals (54 projects, 80.6 %), as well as symptomatic animals (37 projects, 55.2 %), dead carcasses—excluding slaughterhouses—(17 projects, 25.4 %), or animals sampled at the slaughterhouse (29 projects, 43.3 %). Data sources were not mutually exclusive, and one single scenario tree often had diverse data sources.

### 3.3.3. *Reassessment over time*

Martin et al. (2007b) proposed a method to reassess the status of freedom from infection while accounting for historical data and the risk of introduction in between measurements. Thirty-five projects assessing the probability of freedom from infection used this discounting method (Table 4). One of the projects (Foddai et al., 2015) compared two models, with and without temporal discounting. Twenty-four projects (68.6 %) out of the 35 projects that used this temporal discounting used

**Table 3**
General information on the scenario tree model-based projects identified in a scoping review (2006–2021) (n = 67).

| Variable | Values |
|---|---|
| Projects whose (part of) target population were commercially farmed animals | 59 (88.1 %) |
| Projects whose (part of) target population were non-commercially farmed animals | 6 (9.0 %) |
| Projects whose (part of) target population were wild animals | 9 (13.4 %) |
| Projects whose (part of) target population were companion animals | 1 (1.5 %) |
| Projects whose (part of) target population were feral domestic animals | 1 (1.5 %) |
| Projects performed at a national level | 38 (56.7 %) |
| Projects performed at an international level | 6 (9.0 %) |
| Projects performed at a regional level | 16 (23.9 %) |
| Projects performed at the herd level | 6 (9.0 %) |
| Projects which had at least one scenario tree that used only active surveillance data | 31 (46.3 %) |
| Projects which had at least one scenario tree that used only passive surveillance data | 11 (16.4 %) |
| Projects which had at least one scenario tree that used both active and passive surveillance data | 28 (41.8 %) |
| Mean time period considered in the projects in years* | 7.64 |
| Median time period considered in the projects in years* | 5 |
| Minimal time period considered in the projects in years* | 1 |
| Maximal time period considered in the projects in years* | 42 |

* Statistics on time period only comprises projects which had data not originating from a single point in time. When different scenario trees from the same project provided different length, the different scenario trees were kept. This amounted to 46 projects and 50 scenario trees.

**Table 4**
Information on the models used in the different scenario tree model-based projects identified in a scoping review (2006–2021).

| Variable | Values |
|---|---|
| Projects which used a deterministic approach | 1 |
| Projects which used a stochastic approach | 65 |
| Projects which used different components* | 41 |
| Minimum / maximum number of components per scenario tree | 1 / 6 |
| Projects which used risk factors | 50 |
| Minimum / maximum number of risk factors per scenario tree | 0 / 8 |
| Projects which used infection nodes | 67 |
| Minimum / maximum number of infection nodes per scenario tree | 1 / 3 |
| Projects which included diagnostic tests in their scenario tree | 66 |
| Minimum / maximum number of diagnostic tests per scenario tree | 1 / 6 |
| Projects which used detection nodes other than diagnostic tests | 42 |
| Minimum / maximum number of other detection nodes per scenario tree | 0 / 12 |
| Projects which did not reassess the probability of freedom from infection at several time points* | 11 |
| Projects which did reassess the probability of freedom from infection at several time points[†]* | 35 |
| Projects which sampled small populations[§] | 27 |

* as defined in Martin et al. (2007b)
[†]Only projects which assessed the probability of freedom from infection (n = 44) were considered, as the reassessment over time is not usable with only the system sensitivity.
[§]With "small population" are meant populations for which the assumption that the size is infinite and that sampling is done with replacement are not meant. This generally applies to populations with a small number of individuals, but also to larger populations if the sampling frame involves sampling a large number of animals. For the purposes of this scoping review, we have considered the assumption as “not met” if it is either mentioned by the author, or if the data show that more than 10 % of the population was sampled.

at least one model with yearly updating. Nine projects (25.7 %) used at least one model with monthly updating, and three projects (8.6 %) used at least one model with quarterly updating (Ågren et al., 2018; East and Hutchison, 2011; Foddai et al., 2020). One project also used one model with biennial updating (More et al., 2009).

### 3.3.4. Modifications to the original method

The methodology used in the presented models did not diverge widely from the one presented by Martin et al. (2007b). However, we identified projects that brought some modifications or extensions to the original method. Three different projects adapted the method in order to account for imperfect specificity (Fabijan et al., 2019; Nathues et al., 2015; Sergeant et al., 2019). One project introduced the probability of introduction after the calculation of the posterior probability of freedom (Fabijan et al., 2019). Another model accounted for the risk of infection before sampling and during the production cycle (Alba et al., 2017). Another innovation to the original method was the development of a simulation model aiming to evaluate different testing strategies and bio-exclusion practices with simulated data (More et al., 2013). Hansen et al. (2018) included the probability for further transmission if the infection was not detected which allowed for the calculation of the cost of error. Finally, Anderson et al. (2013) innovated, using a virtual grid-cell system as unit for the infection node instead of herds as often used for production animals. This allowed the authors to design a scenario tree adapted to wild animal populations for which the concept of herds is not easy to implement. This adaptation also required the addition of several parameters to the method, such as the probability of capture and the home range of the animals.

## 3.4. Parameters

Fifty-three out of 58 projects published in scientific journals (91.4 %) provided a definition for all the parameters used in the model. Of the 53 projects that described their parameters, 40 (75.5 %) provided the sources of information for all parameters and 50 (94.3 %) provided the value(s) attributed to all parameters. The sources of information for the different parameters can be found in Table 5. This table summarises the information provided in all projects, i.e. also in projects not published in scientific journals.

Values were attributed to parameters either as single values (n = 48, 71.6 %) or as probability distributions (n = 61, 91.0 %). Both were not mutually exclusive—i.e. some parameters of a scenario tree could be defined as single values while others were defined as probability distributions. Risk factors (n = 50) and detection nodes (n = 67) were more

**Table 5**
Sources of information for the parameters used in the scenario tree model-based projects identified in a scoping review (2006–2021) (n = 67).

| Categories | Characteristics | Number of projects which used this category of parameter | Estimation based on published data | Estimation based on laboratory, unpublished data | Estimation based on other unpublished data | Estimation based on an expert elicitation | Estimation based on a risk assessment | Estimation based on international agreements | Estimation based on recommendations from the WOAH | Assumptions made by the authors |
|---|---|---|---|---|---|---|---|---|---|---|
| Risk factors | Definition | 50 | 33 | | 6 | 8 | | 3 | | 13 |
| Risk factors | Weight | 50 | 26 | | 12 | 13 | 3 | | | |
| Detection nodes | Laboratory test sensitivity | 66 | 51 | 5 | 3 | 17 | | | | |
| Detection nodes | Sensitivity of other detection nodes | 42 | 30 | | 7 | 24 | | | | 11 |
| Risk of introduction | | 38 | 16 | | 9 | 4 | 3 | | | 26 |
| Prevalence | Design between-herd prevalence | 67 | 12 | | 2 | 2 | | 20 | 20 | 28 |
| Prevalence | Design within-herd prevalence | 67 | 17 | | 2 | 5 | | 5 | 5 | 26 |

often expressed as probability distributions than as single values. For 34 of 50 (68.0 %) projects some or all risk factors were expressed as probability distributions, while for 27 (54.0 %) some or all were represented by single values. For 54 of 67 (80.6 %) projects, the detection nodes were expressed as probability distributions, versus 24 (35.8 %) that expressed them as single values. This was more balanced for parameters describing the probability of introduction (n = 38), with 22 projects (57.9 %) that expressed them as probability distribution versus 25 (65.8 %) that expressed them as single values.

We found indications that a sensitivity analysis was at least partially performed for 50 of 67 (74.6 %) projects. This amounted to 45 out of 58 (77.6 %) projects when considering only projects that were published in scientific journals.

Of 58 projects published in scientific journals, the authors of 27 projects (46.6 %) fully assessed the confidence of the parameters, while the authors of 30 (51.7 %) projects partially assessed it. For one project, we could not find any information to support the fact that the confidence of the parameters was assessed.

### 3.5. Type of the model

Most projects were developed in Microsoft Excel (n = 44, 65.7 %) and in R (n = 17, 25.4 %) (Fig. 3). However, some authors reported to work with EpiTools (n = 2, 3 %: Jansen et al. 2011 and Welby et al. 2010) or with Python (n = 1.5 %: Anderson et al. 2013, 2017).

Projects implemented in Microsoft Excel often made use of add-ins such as *@RISK* (RRID:SCR_022837), *PopTools* (RRID:SCR_022840) and *ModelRisk* (RRID:SCR_022836). Three projects implemented in R out of 17 (17.6 %) reported the use of the *RSurveillance* (Sergeant 2020, RRID: SCR_021674) package specifically designed for this purpose. The other 14 projects (82.4 %) did not report the use of specific packages for the modelling of STMs. The packages *epiR* (Stevenson and Seargeant (2024), RRID:SCR_021673)—which was merged with *RSurveillance* in 2020—, or *freedom* (Rosendal (2020), RRID:SCR_021675) were never reported as being used. Fig. 3 represents the number of projects that used R or Microsoft Excel through time.

### 3.6. Reporting

#### 3.6.1. Presentation of the results

Thirty projects out of 44 which aimed to assess probability of freedom from infection (68.2 %) provided results as a probability distribution, while only ten (22.7 %) provided results as single values. Of the 30 that used probability distributions, 11 (36.7 %) of them reported a cut-off used to declare freedom from infection. Until now there are neither guidelines nor recommendations to compare a probability distribution to a single cut-off value when substantiating freedom from infection. In total, only seven (63.6 %) of these projects reported how they did this comparison. For two of them, the entire distribution was above the cut-off (Alban et al., 2008; Fabijan et al., 2019). For the comparison of a distribution with the threshold, two projects used the median (Foddai et al., 2020; Lyngstad et al., 2016), two projects used the mean (Mysterud et al., 2020; Norström et al., 2020) and the last project used the 5th percentile (Schuppers et al., 2010).

#### 3.6.2. Quality of reporting and potential methodological flaws

During the data extraction process different reporting problems were identified at multiple levels. The authors often did not mention if the assumptions of the scenario tree methodology were met for their studies. Common assumptions that were neither mentioned nor explained included: a) that the specificity of the diagnostic nodes equals to one; b) independence between tests performed in series; c) no overlapping between surveillance components; and d) sampling in a population of infinite size and performed with replacement. Parameters of the STMs were not always defined, i.e., their values were not always provided, or the sources of data used to attribute values to them were not always

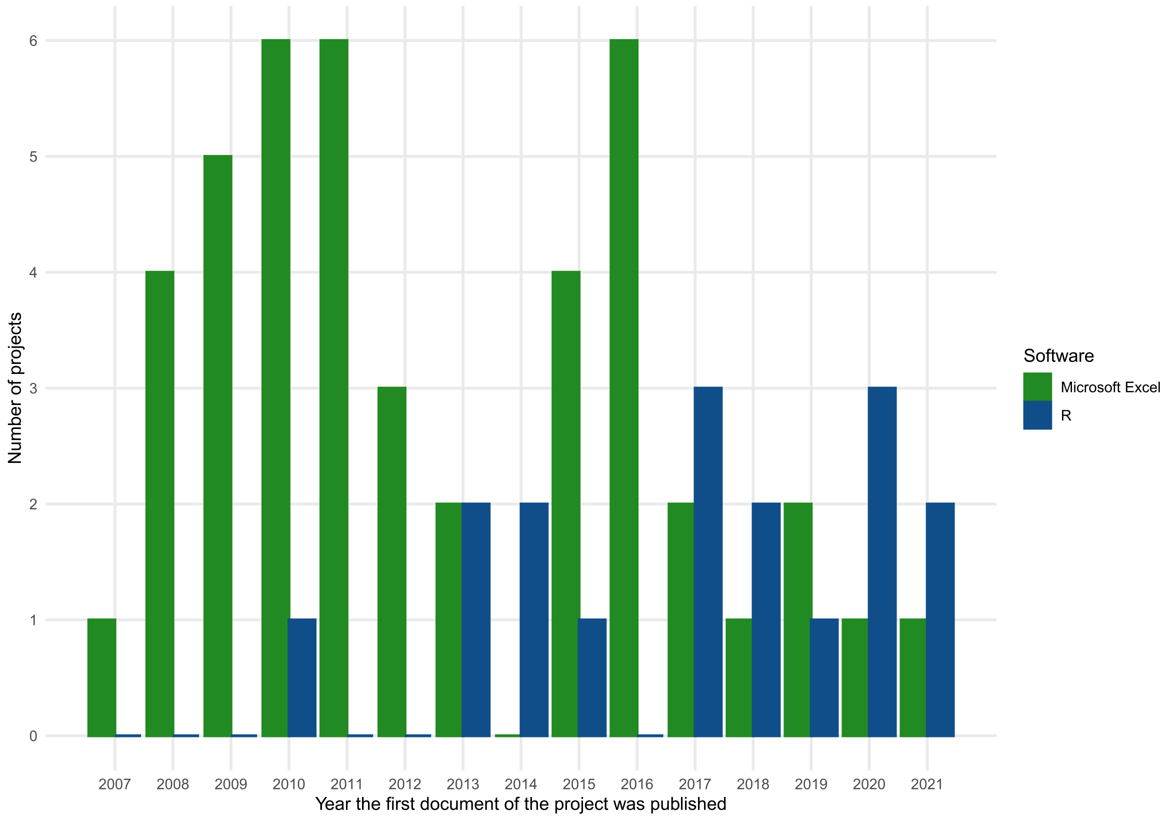


**Fig. 3.** Number of projects which used R or Microsoft Excel per year.

reported. Diverse information about the model used were also commonly missing, such as: a) the use of a different formulas to assess the group sensitivity when the assumptions of infinite group size and sampling with replacement were not met; and b) rounding performed (or not) to calculate the expected number of infected individuals in a group when using the hypergeometric formula to correct for sampling without replacement.

## 4. Discussion

Scenario tree modelling is an important method to support surveillance assessment in the veterinary public health field, allowing a more efficient use of resources by enabling the possibility of combining evidence from different sources of surveillance data and by using risk-based sampling. In this review, we explored and characterized their use, parametrisation, data sources and potential limitations. We also reviewed the reporting of such models.

### 4.1. Use of scenario tree modelling

Scenario tree modelling was used not only to assess freedom from infection, but also to assess sensitivity of surveillance systems for epidemic and endemic infections. In addition, it was used to compare the effectiveness of different surveillance strategies.

Although our results shows that STMs were mostly applied in high-income countries, in particular in Europe, we found that they were also applicable and used in low-income countries. We are uncertain whether this is a consequence of different needs for applying the method between these countries or if it is a consequence of better availability of resources to apply the methods in high-income countries. We also concluded that the methodology is mainly applied to infections affecting farm animals, probably because these populations are economically most relevant. Most studies are conducted at national and, in a smaller proportion, regional level, which is likely linked to the fact that it is mostly implemented by the authorities for trade and/or surveillance purposes.

It is also of interest to note that only a small number of projects used solely passive surveillance to calculate the probability of freedom from infection. Most projects used either active or a combination of active and passive surveillance. This is probably a way for countries to ensure that enough individuals are tested to reach the objectives of the surveillance system. It can also be a strategy to target high risk strata of the population, therefore reducing the number and, consequently, the costs of testing. Legislation requirements can also explain this pattern, as many important infections of livestock are notifiable and therefore a passive surveillance component is already mandatory, which can then be complemented by an active surveillance component to reach the minimum confidence level targeted.

### 4.2. Reporting

Vanderstichel et al. (2013) proposed guidelines for reporting results obtained from STMs and Christensen et al. (2014) proposed guidelines to report updates to a model previously described. However, the extraction process and the analysis of the data extracted emphasised the fact that these guidelines are not widely used to report results of scenario tree modelling. Vanderstichel et al. (2013) suggest reporting the model with seven distinct categories. They present the information in a single table and an additional figure with the graphical presentation of the scenario tree model. STMs can be complex, and the suggested requirements are necessarily detailed. However, the suggested table is complicated, and it may not be easy to adapt the format to the data for all STMs. We hypothesize that these proposed guidelines were not widely adopted as the suggested presentation of the data was both intricate and laborious and that the suggested format for presenting the data did not aid the modelers in presenting the information.

Additionally, during our regular meetings to discuss the data extraction process, we found that the quality of reporting varied between the different projects reviewed that were published in scientific journals. We identified various reporting problems. For instance, we verified that authors commonly fail to mention if the assumptions of scenario trees were met. We also concluded that there was often a lack of detail in the description and discussion of parameters and the model design. In our opinion, this constitutes a hindrance to transparency and the reproducibility of STM models.

Variations in a single parameter can sometimes cause large differences in results, which highlights the need to conduct sensitivity analysis or other assessment on the impact of different parameters on STM outcomes. The interpretation of the results can be strongly affected by the uncertainty attributed to parameters of STMs. However, assessment and/or discussion of the influence of the diverse input parameters on the results was not generalized. Similarly, the uncertainty of values attributed to parameters was often not discussed.

We found that a visual representation of the scenario tree was often useful to understand its design. Furthermore, we found that the publication of formulas—or code—was very effective to understand how calculations were made. However, neither the visual representation nor the code and/or formulas used were reported in every published scientific article.

### 4.3. Misconceptions about sensitivity and specificity

A common misconception we identified relates to the calculation of sensitivity and specificity. Many studies used the sensitivity of the screening test as overall sensitivity of the diagnostic node(s) and assumed a specificity of one, either without any rationale or on the grounds that any positive result would be retested with confirmatory tests to assess whether it really is a true positive. Assuming a specificity of one when the combination of the initial and the confirmatory tests have very high specificity is understandable, but the rationale of how it applies to a particular case should be given. However, the only argument provided by multiple studies was that the assumption is always valid when using the scenario tree methodology, with which we disagree. There are examples where either the pathogen of interest, the field conditions or other constraints can make the diagnostic on live animals challenging, and even serial testing cannot always guarantee a specificity of near 100 % (Alvarez et al., 2012; Viljamaa-Dirks and Heinikainen, 2019). Furthermore, confirmatory tests often do not have a perfect sensitivity. Therefore, we think that their influence on the overall sensitivity of the diagnostic nodes should be taken into consideration and discussed.

Correlation in serial testing was rarely discussed in the reviewed documents. However, ignoring it implies assuming independence of the tests, which is often not true. Failing to take lack of independence between the tests into account can lead to incorrect estimation of the overall sensitivity (Georgiadis et al., 2003).

It is important to mention that, in some STMs, the outcome of the model is not sensitive for changes the diagnostic sensitivity. In such cases, it can be acceptable to simplify the model by assuming independence of the tests or by ignoring potential confirmatory tests. However, we think that this decision should be clearly stated and discussed when reporting outcomes of such STMs.

Detection nodes differed greatly between different scenario trees. Considering this complex framework and the gaps highlighted on the attribution of values to sensitivity and specificity parameters, we think it is important to increase awareness of authors about the importance of these nodes when designing a scenario tree.

### 4.4. Calculation of group sensitivity

The basic methodology assumes that the sampling population is of infinite size and that sampling is performed with replacement. However, these assumptions are not always met. Martin et al. (2007b) suggested the use of binomial approximation of the hypergeometric distribution proposed by MacDiarmid (1987) to calculate the sensitivity in such contexts. The number of projects containing populations for which the assumption of sampling with replacement on a population of infinite size were not met is given in Table 4. However, this information was often not reported, neither explicitly nor by providing the formula used, and had to be inferred by the reviewers according to other details provided in the documents. Furthermore, the formulas used to calculate sensitivity for sampled populations were seldom reported. When this information was provided, the hypergeometric distribution was the method. However, authors rarely reported whether the exact hypergeometric distribution or its binomial approximation was used.

When using the exact hypergeometric distribution, the expected number of positive individuals (expressed as an integer) should be used, instead of the prevalence. Its binomial approximation can mathematically work with a real number. Martin et al. (2007b) proposed to calculate the expected number of positive individuals by multiplying the total number of individuals by the prevalence. This value is then rounded up, as from a biological point of view, there are no "partly infected" individuals. This approach is used by default in the R packages *RSurveillance* (Sergeant 2020, RRID:SCR_021674) and *epiR* (Stevenson and Sergeant 2024, RRID:SCR_021673). The original and default behaviour of the R package *freedom* (Rosendal (2020), RRID:SCR_021675) is not to round up the expected value to avoid potential overestimation of the group sensitivity, and therefore accept "partly infected" animals in the model. However, since the package was first made available, options for alternative rounding rules were added, though at time of writing (2024–02–26), these adaptations are only available on the GitHub version of the package. The Swiss Federal Food Safety and Veterinary Office used an intermediate approach in several surveillance programmes. The expected value is rounded to one if it would be below one, with the argument that an infected group must contain at least one infected individual to be infected; the value is not rounded if it lies above one, to avoid an overestimation of the group's sensitivity (G. Delalay, Federal Food Safety and Veterinary Office, Switzerland, personal communication). We tried to extract information on the rounding method used from the documents, but it was almost never clearly stated, and therefore we could not include it in this review.

### 4.5. Interpretation of the results

One of the primary objectives of the scenario tree methodology is to provide evidence of freedom from infection for trade partners. The outputs of STMs might be among the pieces of evidence required to recognize freedom from infection. In this context, international organisations like WOAH and EU often require a level of confidence needed for trade partners to accept that the infection prevalence in the country or region lies below the design prevalence. Most of the time, this level of confidence is provided in form of a single value that may be considered as a cut-off. However, most studies are made in a stochastic way and their results often are expressed as probability distributions. There is for now no internationally accepted method to compare a probability distribution with a single cut-off value. This lack of guidance is a relevant problem, as it can also get quite complex to conceptually grasp what the results of such studies—when expressed as distributions—really represent. When stochastic studies did compare their results to the level of confidence expressed as a cut-off value, they did it in heterogeneous ways. Some of them used the median, others the mean and the remaining the fifth percentile of their probability distribution. To our knowledge, no author did calculate the combined probability mass of those distributions to compare it to the cut-off. However, the reviewers made the observation that when plotting graphs or providing single values instead of the probability distributions obtained with the model, authors often tended to use the median. Because an international agreement on this question is missing, it was not clear for most documents which conclusions should be taken at the end of the study. Therefore, we strongly emphasise the need for international agreement to address this question.

### 4.6. Outline of the scenario tree

We have observed discrepancies between the projects on how the definitions provided in the original article from Martin et al. (2007b) were understood, mainly concerning the different node types.

The ability to use different data sources or surveillance strategies and to combine them together with the use of different components is highlighted by Martin et al. (2007b) as the biggest advantage of the method. Indeed, we observed that more than half of the projects used different components.

Risk factors were also frequently used. The targeting of high-risk strata is an efficient way for governments to achieve high sensitivity with lesser costs due to testing fewer herds or animals, as opposed to representative sampling of the population (Hadorn et al., 2002).

There was a high variability between the studies regarding the different nodes used and how they were combined. The use of detection nodes is a good example of this variability. They reflected the different surveillance strategies, ranging from laboratory diagnostic tests on samples actively collected from randomly selected animals to complex strategies with diverse selection criteria based, for instance, on symptoms or recognition of lesions. The high variability of the outline of the scenario tree indicates that the methodology is highly flexible and can be adapted to a large variety of surveillance systems. On the other hand, it also means that the authors should pay attention when drawing the whole tree, particularly the different detection nodes, so that it accurately represents the surveillance strategy they want to model.

Most of the projects used the temporal discounting method proposed by Martin et al. (2007b). This allows to use historical data in the model and therefore allows for the reduction of the sampling size and costs associated with testing in continuously conducted monitoring and surveillance programmes. However, it must be noted that this temporal discounting is only relevant for models that estimate probability of freedom from infection, and not for models that only estimate the sensitivity of the surveillance programme.

### 4.7. Accessibility of scenario tree models

Results show that most of the projects included in the study were published in scientific journals. However, the method was originally developed to allow countries to prove freedom from infection to their trading partners. Thus, we expected a high number of non-scientific publications. This led us to look specifically for grey literature during our search. We obtained a low number of non-scientific reports: six projects with material provided as a report, with one project (Doherr, 2016) being an internal report for the government, which is not publicly available. We believe the predominance of scientific manuscripts in our results is linked to the difficulties to obtain relevant grey literature. In addition, it is likely that such reports are published in native languages, which might differ from English, further compromising their retrieval. Scientific journals usually do not publish manuscripts on routine surveillance programmes, as a common requirement to be published is to report something new. However, as governments generally only adapt an already existing method, it can be hard for them to find a scientific journal where to publish it if there is no change done to the original method. Additionally, in such cases the STMs are made for decision support and in many cases national funding bodies do not see the merit in scientific publication. The issue of the availability of data routinely collected for animal or human health surveillance is not new and was already raised (Edelstein et al., 2018).

## 5. Conclusion

Scenario tree modelling is a pivotal and versatile methodology in the field of animal infectious disease surveillance and, over the years, it has been extended to other fields of veterinary public health, such as assessments of animal welfare and antimicrobial residues. However, the lack of standardization when reporting results makes the understanding of the models difficult and hinders their reproducibility and comparability. We also highlighted common problems and misconceptions in the design and parameterisation of STMs. Furthermore, many studies using STMs often lack sufficient detail on how the results should be interpreted and concretely applied to substantiate freedom from infection. Therefore, we stress the need for good documentation and training opportunities relating to the STM methodology, as well as internationally accepted reporting guidelines. Such guidelines would allow for better study reproducibility, more adequate comparisons between STMs and more accurate and complete interpretation of studies' results. More comprehensive reporting would also help readers and reviewers to identify potential weaknesses in the design of STMs or, otherwise, to understand why unintuitive assumptions might be correct in a given context.

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## CRediT authorship contribution statement

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## Declaration of Competing Interest

None to declare. However, some of the authors and their respective teams have worked or still work with scenario tree models. The authors did not extract data from papers that they co-authored. The funder did not take part in the development or execution of the project.

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## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.prevetmed.2024.106371.

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